Are OT-Aligned Flows What You Need?

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Abstract & Motivation

We study how the geometry of continuous-time generative models affects both sampling efficiency and quality on image generation and robot action generation tasks. Generative modeling is realized as a probability path $(p_t)_{t\in[0,1]}$ pushing a simple base distribution toward the data distribution.

We compare four paradigms under matched U-Net architectures and training pipelines:

- **Diffusion Models [1, 2]:** Curved SDE paths with no sample coupling.
- **Flow Matching (FM) [3]:** Gaussian OT interpolants with no sample coupling.
- **Rectified Flow (RF) [4]:** Straight-line Gaussian interpolants with independent coupling.
- **Optimal Transport CFM (OT-CFM) [5]:** Gaussian OT interpolants with minibatch OT couplings to approximate W_2 geodesics.

Methodology: Continuous Normalizing Flows

A flow ϕ_t is defined as the solution to the neural ODE $\frac{d}{dt}\phi_t(x) = v_t(\phi_t(x))$ [6]. Training is performed via the simulation-free **Conditional Flow Matching (CFM)** objective:

$$\mathcal{L}_{CFM}(\theta) = \mathbb{E}_{t,x_1,x\sim p_t(\cdot|x_1)}||v_t(x;\theta) - u_t(x|x_1)||^2$$

The Geometric Comparison

Model	Path Geometry	Coupling Strategy
Diffusion	Curved	No pairing
FM	OT Gaussian Interpolant	No pairing
RF	OT Gaussian Interpolant	Independent
OT-CFM	OT Gaussian Interpolant	Minibatch OT (W_2)

Table 1. Comparison of generative paradigms. FM and RF use conditional OT Gaussian interpolants for training, while OT-CFM additionally enforces minibatch OT couplings to approximate the full optimal transport problem.

Key Finding: Geometry vs. Quality

OT-CFM learns the straightest and lowest-energy flows in our CIFAR-10 experiments, yet offers no FID improvements over the simpler FM/RF paths. In robotics, CFM-based policies achieve strong one-shot and OOD performance and produce straighter, lower-energy denoising trajectories than Diffusion Policy. However, OT-CFM itself collapses in this setting.

Experiment I: Image Generation (CIFAR-10)

We evaluated FID, path straightness, and Benamou-Brenier kinetic energy on CIFAR-10.

Model	FID ↓	Straightness ↑	Kinetic Energy ↓	NFE
Diffusion (Stoch.)	37.46	-	-	1000
Diffusion (Flow)	311.24	0.9468	$7.10e+01$	99
FM w/ OT	8.78	0.9776	$5.87e+03$	194
Rectified Flow (RF)	**8.76**	0.9779	$5.87e+03$	194
OT-CFM	9.04	**0.9847**	**5.83e+03**	188

Table 2. Results on CIFAR-10. NFEs for stochastic diffusion correspond to the fixed sampling schedule, while NFEs for all ODE-based samplers reflect the number of function evaluations used by an adaptive RK45/Dopri5 integrator with fixed absolute and relative tolerance (10^{-5}).

- OT-CFM learns the straightest, lowest-energy flows, confirming it approximates the W_2 geodesic.
- However, it does **not** outperform simpler FM/RF paths in sample quality (FID).
- We hypothesize that the OT objective is **under-optimized** at our batch size (128)
- Consistent with prior literature, we observe that flow-based models form images early, while diffusion models remain noisy until the final steps, see Figure 1.

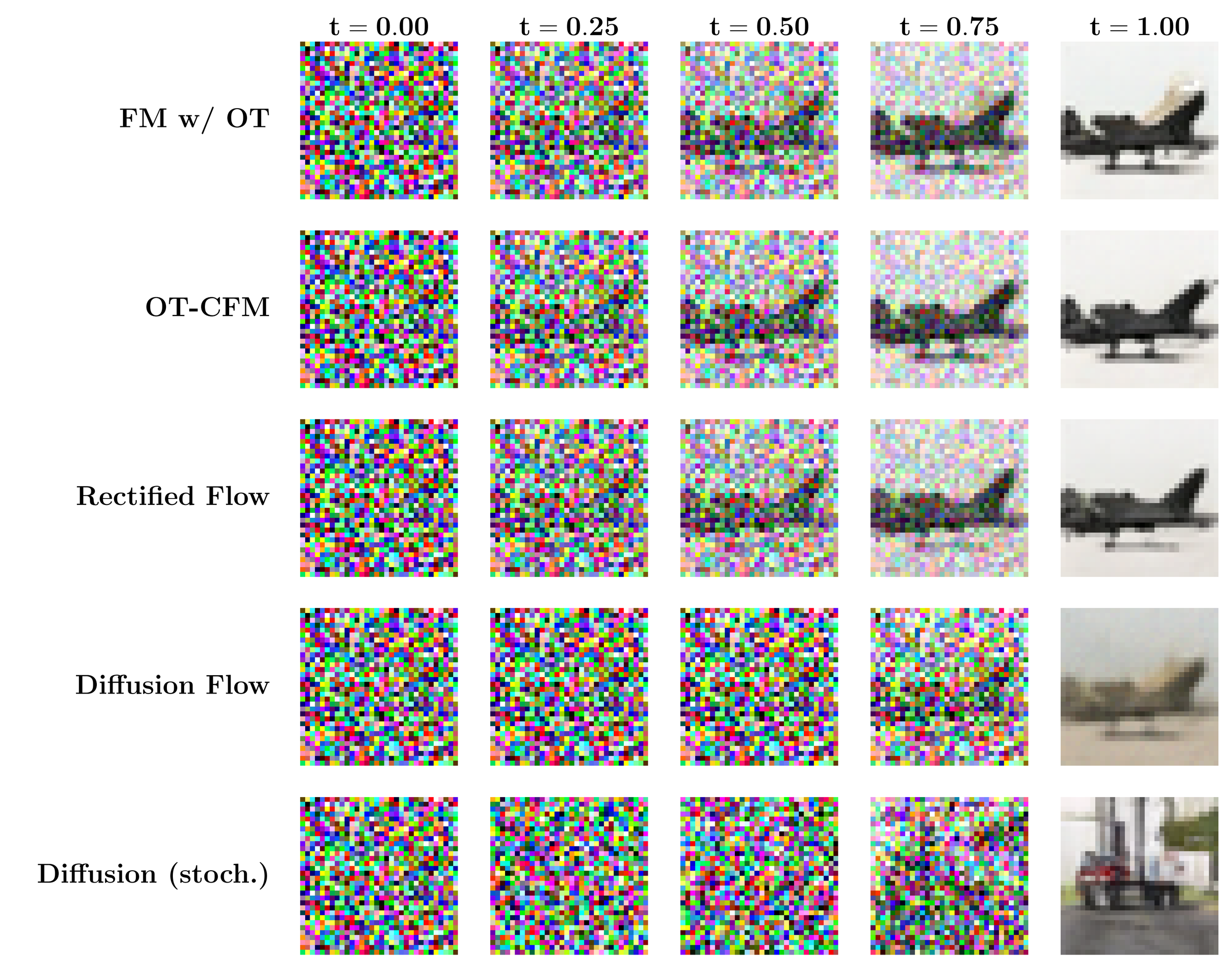

Figure 1. Comparison of flow-based and diffusion-based noise-to-image trajectories on CIFAR-10.

Experiment II: Robot Action Generation

We compare Diffusion Policy (DP) [7] against CFM-based policies on a **cube-stacking task** using a Panda robot in the ManiSkill simulator.

- **Task:** Pick a red cube and place it on a green cube.
- **OOD Generalization:** An obstacle is inserted between cubes, requiring a detour.

Visualizing Trajectories: We find that CFM-based policies yield straighter denoising trajectories with lower energies compared to DP, see Fig. 2.

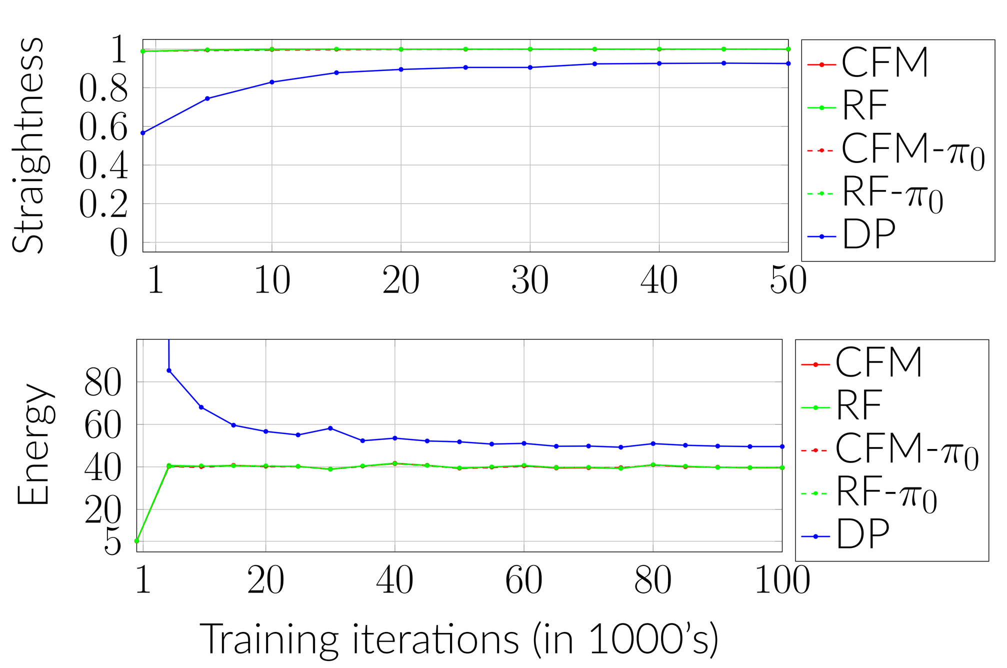

Figure 2. Evolution of the straightness (top) and energy (bottom) of the integration paths throughout training.

Key Result: One-Shot Action Generation

- Diffusion Policy collapses under single-step inference (0% success).
- CFM-based policies achieve high success rates with a **single function evaluation**, both for in-distribution and OOD tasks. See Fig. 3 for an OOD trajectory generated by FM policy.
- $FM - \pi_0$ reaches 85% success with 1 step.
- For larger integration steps, DP and CFM policies show similar performance and similar generalization-scaling trends.

Table 3. Performance across ID and OOD settings with varying number of inference steps.

	ID			OOD		
Method	1-shot	10-shot	50-shot	1-shot	10-shot	50-shot
FM	0.23	0.88	0.95	0.46	0.67	0.80
RF	0.61	0.94	**0.98**	0.41	0.67	0.77
FM-π_0	**0.85**	0.94	0.95	**0.51**	0.64	0.67
RF-π_0	0.60	0.91	0.94	0.45	**0.74**	0.66
OT-CFM	0.0	0.0	0.0	0.0	0.0	0.0
DP	0.0	**0.99**	0.97	0.0	**0.74**	**0.92**

Why? CFM models learn nearly linear paths ($Straightness \approx 1$), whereas Diffusion Policy learns curved paths that require iterative refinement.

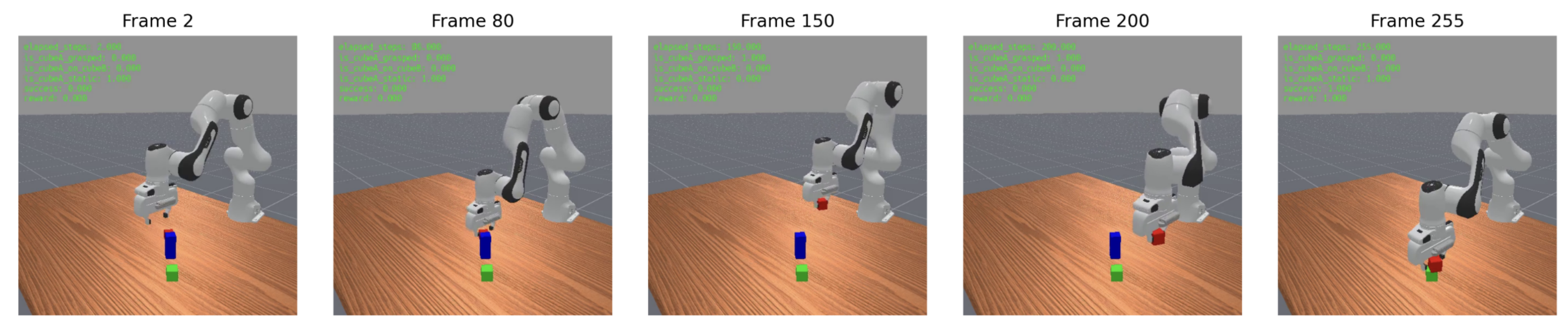

Figure 3. Robot trajectory in the OOD task generated by the FM-based policy.

Discussion & Conclusion

- **Geometry Matters for Control:** The "straightness" of the learned flow correlates strongly with 1-shot success in robotics.
- **Efficiency:** CFM-based policies enable substantial step-size coarsening without catastrophic failure, a robustness property that DP does not exhibit. For larger integration steps, CFM and DP exhibit similar performance.
- **Optimization Trade-offs:** On CIFAR-10, OT-CFM reduces kinetic energy but does not improve FID, suggesting robust straight-line paths (RF) are sufficient for image quality.

Future Work: Investigate larger batch sizes for OT-CFM to reduce coupling noise and apply CFM policies to higher-frequency control loops on hardware.

References

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Code at github.com/mletey/CS2840-ot-aligned-flows and github.com/JackBJ23/cfm-policies